

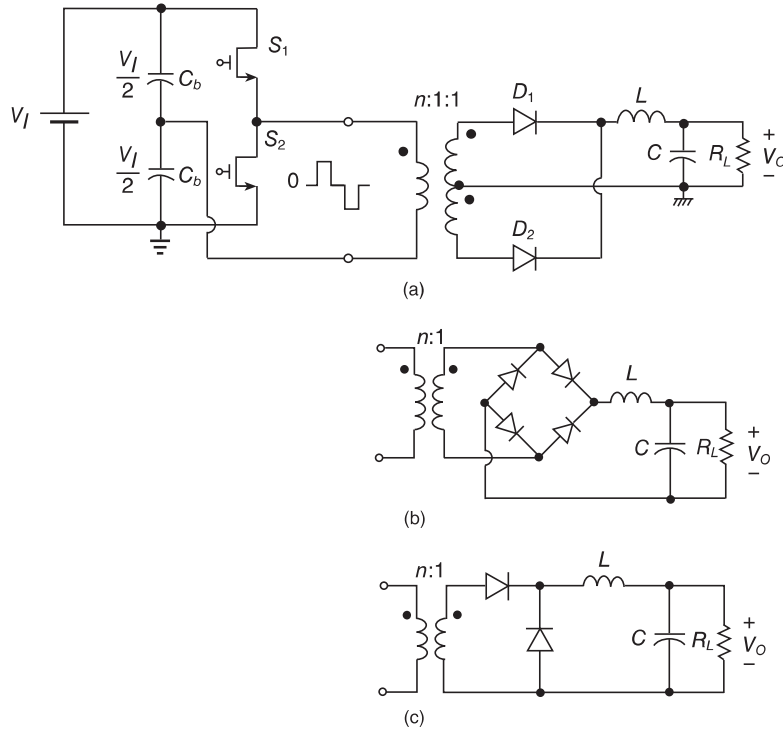


Figure 7.1 Half-bridge converter with two blocking capacitors. (a) With a transformer center-tapped rectifier. (b) With a full-bridge rectifier. (c) With a half-wave rectifier.

the same time. In this case, the dc input voltage V_I is connected to the ground through two on-resistances of the MOSFETs, generating a very large current spike and destroying the transistors. For example, if dc input voltage is $V_I = 340$ V and the MOSFET on-resistance is $r_{DS} = 1 \Omega$, then the peak current through the MOSFETs is $I_{pk} = V_I / (2r_{DS}) = 340/2 = 170$ A. The phenomenon when both transistors are conducting at the same time is called a *cross conduction*, and the current through the transistors is called a *shoot-through current*. The switching network of the inverter has two transistors in series (a totem pole arrangement), resulting in difficulty in driving the high-end (upper) transistor because its gate is not referenced to ground. A small wide bandwidth (pulse) transformer is normally used to drive the upper transistor to achieve a square-wave gate-to-source voltage. A second pulse transformer is usually added to drive the bottom transistor. This driver provides dc isolation in the control path. DC isolation is required in the power stage and the control circuit to accomplish a dc isolation between the input and the output of a dc–dc converter. Pulse transformers also provide protection of a control circuit against high-voltage breakdown. The transistor may be broken in such a way that there is a short circuit between the transistor drain and gate, and a high voltage V_I appears at the transistor gate. If the control circuit is directly coupled to the gate, a high voltage is applied across the control circuit output, destroying the control circuit. However, if a control circuit is coupled to the gate through a pulse transformer, the control circuit will not be damaged by a high voltage at the transistor gate.

Assuming that both blocking capacitors C_b are identical, the voltage drop across each of them is $V_I/2$. These are usually large electrolytic capacitors with very large tolerances, for example, -20% and $+100\%$; therefore, the voltage drops across them may be different from $V_I/2$. To make these voltages closer to each other, large balance resistors R_b of the order of $100 \text{ k}\Omega$ to $1 \text{ M}\Omega$ may be connected in parallel with the blocking capacitors.